Spiders Vs Bees

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Introduction

Exchange-Traded Funds (ETFs) and their flagship product, Nifty BeES, have emerged as pivotal instruments in India's investment landscape, offering retail and institutional investors a low-cost, diversified gateway to benchmark indices like the Nifty 50. Launched in December 2001 by Benchmark Mutual Fund, Nifty BeES—short for Benchmark Exchange Traded Scheme—marked India's entry into passive investing, tracking the performance of the top 50 blue-chip companies with minimal expense ratios and high liquidity through stock exchange trading. Their rapid ascent was fueled by key milestones: a 2015 boost from the Employees' Provident Fund Organisation's (EPFO) equity allocations via ETFs, SEBI's 2018 ban on upfront mutual fund commissions that leveled the playing field for passive funds, and government-backed initiatives like the Bharat-22 ETF for divestment proceeds. From a modest Rs 2 crore AUM in 2002, Nifty BeES alone ballooned to Rs 10,000 crore by 2022, while the overall Indian ETF market achieved a staggering 45.9% CAGR over the past decade, democratizing market access amid surging retail participation post-pandemic.

There are three major reasons why this analysis of Nifty BeES is important for the Indian market. First, the principal advantage of BeES over traditional index mutual funds is that they can be purchased and sold at real-time prices throughout the trading day on the NSE. As we will demonstrate, low-cost index funds still deliver marginally higher returns than BeES due to tracking differences and expense ratios. Given that investors can freely choose either vehicle, this return differential quantifies the implicit cost of intraday liquidity—or the value of immediacy—in the Indian context, a key concept in market microstructure studies. Second, Nifty BeES has evolved into a cornerstone investment vehicle, commanding over 10,000 crore in AUM and significant daily trading volume, making its performance and behavior critical for investors, regulators, and market participants. Third, the ETF structure pioneered by BeES is now widely regarded as the blueprint for the future of passive investing in India, with SEBI and the government promoting similar products for divestment and retail participation; thus, understanding both its returns and the impact of its exchange-traded organizational form is essential.

Among the prominent ETFs tracking the Nifty 50 Index—such as ICICI Prudential Nifty ETF, UTI Nifty ETF, and Nippon India ETF Nifty 50 BeES—we aimed to select one that most closely aligns with the structural and functional characteristics of the SPDR S&P 500 ETF (SPY) used in Elton et al. (2002). For our replication analysis, the ETF needed to possess a long trading history, high liquidity, low tracking error, and a transparent in-kind creation and redemption mechanism similar to that of SPDRs. Among the available options, Nippon India ETF Nifty 50 BeES best satisfies these criteria. Launched in 2002 as India's first ETF, it has maintained consistent replication of the Nifty 50 Index, exhibits one of the lowest tracking errors, and benefits from robust secondary market activity. Its long operational history and structural similarity to SPDR make it the most appropriate choice for analyzing index replication and performance in the Indian context.

Before analyzing Nifty BeES, we will briefly review their history and important characteristics. Each BeES unit represents an ownership interest in the Nippon India ETF Nifty 50 BeES scheme. The scheme, as stated in the prospectus, holds all of the common stocks in the Nifty 50 Index and is intended to provide investment results that, before expenses, generally correspond to the total return performance of the Nifty 50 Total Return Index. BeES units are traded on the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE) and can be bought and sold like any stock at any time during the trading day. One BeES unit has a price equal to approximately one-tenth of the price of the Nifty 50 Index. The initial deposit creating BeES units was made on December 28, 2001. The scheme was organized as an open-ended exchange-traded fund under the SEBI (Mutual Funds) Regulations, 1996, with no fixed

mandatory termination date, allowing for perpetual operation subject to regulatory approvals and scheme provisions. Any such scheme is governed by a trust agreement under the Indian Trusts Act, 1882, and there are certain aspects of the trust agreement governing BeES that are important to understand. First, BeES charges an expense fee to holders of the units. This has historically been 4 basis points per annum. Second, a specific mechanism exists for changing the number of BeES units outstanding. Authorized participants can create or redeem BeES units in minimum creation unit sizes of 50,000 units by engaging in in-kind transactions plus getting or receiving certain sums of cash. For example, participants can turn in a basket of stocks matching the Nifty 50 Index in the same proportion, plus cash equal to the accumulated dividends less management expenses, and receive BeES units in return. Participants can do so for a nominal transaction fee as prescribed by SEBI guidelines (typically low and fixed, regardless of the size of the transaction).

Unlike the SPDR Trust, which accumulates dividends in a non-earning account and pays them out quarterly, Nippon India ETF Nifty 50 BeES reinvests dividends in real time as they are received from the underlying Nifty 50 stocks. These reinvested dividends are immediately used to purchase additional units of the constituent stocks in proportion to the index, ensuring the scheme closely tracks the Nifty 50 Total Return Index (TRI). As a result, no cash dividends are paid out to unitholders; instead, the benefit of dividend income is reflected in the daily NAV growth, enhancing compounding and aligning performance with the total return version of the benchmark.

Having provided background on Nippon India ETF Nifty 50 BeES, we turn to the purpose of this analysis: to study the performance of Nippon BeES and to compare it with other methods of passive indexing in the Indian market. This analysis proceeds as follows. In Section 1, we examine the performance of Nippon BeES as an investment vehicle. We start by comparing the return from holding Nippon BeES units with the return from the Nifty 50 Total Return Index (TRI). In this section, we first evaluate Nippon BeES returns as if units could be bought and sold at their indicative net asset value (iNAV). We then examine the magnitude and time path of the differences between Nippon BeES market price and iNAV (commonly referred to as premium/discount). Since Nippon BeES is not the only way to gain exposure to the Nifty 50, we next compare the return on Nippon BeES with the return on other indexing vehicles: traditional index mutual funds and Nifty 50 futures. Since creation and redemption data for Nippon India ETF Nifty 50 BeES is available only to authorized participants and not publicly disclosed, we will skip the detailed analysis of the creation and redemption process—unlike the approach taken in the seminal paper "Spiders: Where Are the Bugs?" by Elton et al. (2002), which leveraged such data to examine arbitrage efficiency and tracking errors in the SPDR S&P 500 ETF.

In the final section, we examine the determinants of trading volume in Nippon BeES. The drivers of volume offer valuable insight into who is trading Nippon BeES—whether retail investors, institutions, or authorized participants—and why they are using this ETF, particularly in the context of intraday liquidity, index arbitrage, portfolio rebalancing, or speculative positioning in the Indian market.

1 Performance of BeES

The purpose of this section is to examine the return on Nippon BeES to determine if it serves as a reasonable investment instrument. Since the basic value of Nippon BeES is derived from the Nifty 50 Index, we compare the return on holding Nippon BeES units with the return on the Nifty 50 Total Return Index (TRI) and decompose any differences to identify their sources. We break the Nippon BeES return into two components: the return due to changes in the net asset value (NAV) and the return due to deviations between market price and NAV (premium/discount). This decomposition isolates performance from temporary mispricings. Over long holding periods, arbitrage via creation and redemption by authorized participants ensures that price-NAV deviations remain small and transient. For instance, over our sample period from 2013 to 2021, the average annualized return from holding Nippon BeES was **14.23%**, while the return on NAV was also **14.23%**, indicating near-perfect alignment after accounting for expenses and tracking differences. After examining overall return, we will analyze the factors causing any divergence between Nippon BeES returns and the Nifty 50 TRI. Later, we will compare Nippon BeES performance with that of other Nifty-linked instruments, such as index mutual funds and futures.

1.1 Overall Returns on BeES

We begin by calculating the return an investor would earn from Nippon BeES if units were always bought and sold at Net Asset Value (NAV)—eliminating any impact from market price premiums or discounts.

Table 1: Nifty BeES Annualized Return Decomposition (2013-2021)

Component	**Annualized bps**
NAV Return Underperformance (vs. TRI)	−0.53
Tracking Error (Price-based vs. Price Index)	−1.20
Residual Component (Expenses & Dividends)	+0.66

*Total shortfall of −0.53 bps is the sum of Tracking Error (−1.20 bps) and Residual Component (+0.66 bps).

The NAV of Nippon BeES reflects the market value of the securities constituting the Nifty 50 Index, plus accumulated dividends on the underlying shares, net of management expenses. For the purpose of this study, we also constructed an adjusted NAV series excluding dividend and management fee components to isolate the pure price-based return.

We report the NAV return from holding Nifty BeES for each year of the sample period. Since our objective is to assess total return, we computed the monthly returns as the change in NAV plus the dividends distributed to ETF holders, divided by NAV, and then compounded them to obtain annualized values. Dividend payments were assumed to be reinvested in the ETF on the respective payout dates.

The first step was to compare these returns with the returns on the Nifty 50 Total Return Index (TRI), which captures both price appreciation and reinvested dividends of the underlying stocks. To construct a dividend-adjusted benchmark, we used both the Nifty 50 Price Index and the TRI series, enabling us to decompose total return differences into price-based tracking discrepancies and the effects of dividend accrual and management fees. Unlike Elton et al. (2002), who used CRSP data as an alternative to the Standard & Poor's index values, we relied on NSE-reported Nifty trading data and the TRI series to ensure consistency with the Indian market framework.

When we examine the return on Nifty BeES relative to the Nifty 50 TRI, we find that, on average, the NAV return underperforms the index by 0.53 basis points per year. This indicates that, over the sample period, ETF investors earned slightly lower total returns than those implied by the benchmark index. When we isolate the price-based component by excluding dividends and management fees from the NAV, the tracking error—representing the difference between the ETF's underlying portfolio return and the Nifty 50 Price Index return—averages −1.20 basis points per year. This reflects minor underperformance due to replication frictions and portfolio adjustment costs.

Interestingly, the residual component, representing the effect of expenses and dividends, equals +0.66 basis points per year. This positive contribution suggests that, despite minor price-based underperformance, efficient dividend management and lower expense ratios of Nifty BeES offset part of the tracking loss. This finding contrasts with Elton et al. (2002), where the analogous component was negative, largely due to the SPDR trust's non–interest-bearing dividend holdings and higher management fees.

In summary, while Nifty BeES underperforms the Nifty 50 Index marginally on a total return basis, the decomposition reveals that the shortfall is primarily due to minimal tracking inefficiencies rather than cost structure. The positive expenses and dividends effect highlights the improved efficiency of Indian ETFs relative to early-generation SPDRs, driven by more active dividend distribution mechanisms and lower expense ratios.

Tracking Error

The difference in performance due to tracking error is straightforward to estimate. By taking the NAV of Nifty BeES and subtracting the accumulation component—which represents dividends received from the underlying securities but not yet paid out, adjusted for management fees—we obtain a measure of return based solely on price movements of the ETF's portfolio. This price-based NAV return can be directly compared with the return on the Nifty 50 Price Index, as both exclude dividend reinvestment effects. This comparison allows us to evaluate how closely Nifty BeES replicates the underlying index before accounting for income and expenses.

When we compare the price-only NAV return of Nifty BeES with the Nifty 50 Price Index, we find a tracking error of −1.20 basis points per year, indicating a small degree of underperformance. This shortfall reflects minor replication frictions such as rebalancing delays, transaction costs associated with index adjustments, and minimal deviations between the ETF's holdings and the actual index composition. However, given the ETF's passive management approach and low turnover, these effects are economically insignificant. The small magnitude of the tracking error suggests that the fund's portfolio management

closely follows the index methodology, and that replication inaccuracies play a negligible role in explaining total underperformance.

Unlike Elton et al. (2002), where the authors compared Spider returns against both the Standard & Poor's Index and the CRSP Index to assess index construction discrepancies, our analysis relies solely on NSE-reported Nifty data. The Nifty 50 Price Index represents the official market benchmark, while the TRI captures the reinvested-dividend version of the same index, eliminating the need for a secondary data source such as CRSP. Consequently, our framework ensures consistency between ETF returns, index prices, and dividend accruals within a unified domestic data environment.

When we aggregate our results, the total shortfall of -0.53 basis points can be decomposed into the tracking component of -1.20 basis points and a positive expense-and-dividend adjustment of $+0.66$ basis points. This decomposition reveals that, although the ETF slightly underperformed the index on a pure price basis, efficient dividend reinvestment practices and a lower expense ratio offset part of this difference. Overall, the Nifty BeES demonstrates exceptionally tight tracking relative to the benchmark, with deviations well within the range of random short-term fluctuations rather than structural inefficiencies.

1.2 Comparison with Alternative Index Funds

An individual investor in India has several ETF or index-fund options designed to replicate the Nifty 50 Index. Among the most prominent are Nippon India ETF Nifty 50 BeES, UTI Nifty 50 ETF, Nippon India Nifty 50 Index Fund, and ICICI Prudential Nifty Index Fund.

To evaluate how Nifty BeES performs relative to these alternatives, we regressed the daily returns of each fund on the daily returns of the Nifty 50 Total Return Index (TRI).

The estimated intercepts are close to zero for all funds, ranging between 3.17×10^{-5} and 4.62×10^{-5}, while the corresponding t-statistics are highly significant (around 4–5).

The slope coefficients on the Nifty TRI return are all near unity—0.998 for Nifty BeES, 0.995 for UTI, 0.989 for Nippon, and 0.989 for ICICI Prudential—with exceptionally high t-values (ranging from 747 to 1244).

These results confirm that all four instruments track the Nifty 50 Index very closely, with Nifty BeES showing the coefficient closest to one, indicating the tightest replication of index movements.

Across funds, the small positive intercepts suggest minimal daily excess returns, most likely attributable to rounding and dividend-timing effects.

Overall, these regressions imply that differences in performance across Indian index funds arise primarily from cost structures and portfolio-management practices, rather than from systematic deviations in index replication.

Given Nifty BeES's high liquidity, long operational history, and almost perfect tracking coefficient, we use it as the principal ETF for further analysis.

We've verified that tracking error and data-source bias aren't the reason for the shortfall. Therefore, the residual we analyze next really is due to costs and dividends.

1.3 Deviation of Price from NAV

An important aspect of evaluating exchange-traded funds is to examine how closely the market price of the ETF follows its Net Asset Value (NAV). For an ETF structured like Nifty BeES, the NAV represents the current market value of the underlying securities plus accrued dividends, net of management fees. The ETF's market price, however, is determined by supply and demand in the secondary market and may temporarily deviate from its NAV due to liquidity conditions, market sentiment, or short-term frictions in the creation and redemption process. To analyze this relationship, we calculated the daily proportional deviation defined as $\left(\frac{\text{Price}}{\text{NAV}}\right) - 1$, and examined its persistence over time.

To test whether these deviations persist or mean-revert, we regressed the daily deviation at time $t+1$ on the deviation at time t:

$$\left(\frac{\text{Price}}{\text{NAV}} - 1\right)_{t+1} = \alpha + \beta \left(\frac{\text{Price}}{\text{NAV}} - 1\right)_{t} + \varepsilon_t$$

The estimated regression yields an intercept of -0.0108 ($t = -2.20$) and a slope coefficient of 0.6305 ($t = 38.72$), with an $R^2 = 0.397$. The statistically significant negative intercept suggests that, on average, deviations tend to decline over time rather than persist. The slope coefficient being less than unity indicates a strong mean-reverting pattern in price deviations: when Nifty BeES trades at a premium (or discount) relative to NAV on one day, approximately 63% of that deviation carries over to the next

trading day, with the remainder corrected through market mechanisms such as arbitrage or authorized participant activity.

The magnitude of R^2 (≈ 0.40) implies that almost 40% of next-day variation in price deviations can be explained by current-day deviations, reflecting moderate short-term persistence but rapid correction. This behavior is consistent with the creation–redemption structure of ETFs, which allows arbitrageurs to align the market price with the NAV whenever significant premiums or discounts arise. The negative intercept also indicates that the long-run expected deviation is close to zero, consistent with efficient pricing in the ETF market.

Overall, these results show that Nifty BeES trades very close to its NAV, and any temporary divergences are small, short-lived, and corrected quickly through arbitrage activity. This finding parallels Elton et al. (2002), who observed similar mean-reverting deviations in SPDRs. However, the slightly lower persistence observed in our results ($\beta = 0.63$ compared with SPDRs' ≈ 0.80) suggests that the Indian ETF market exhibits somewhat faster correction dynamics, possibly due to narrower bid–ask spreads and efficient participation by institutional arbitrageurs on the NSE.

An R^2 of 0.40 indicates that nearly 40% of today's deviation in Nifty BeES price from its NAV carries over to the next trading day, suggesting that price–NAV gaps in the Indian ETF market persist longer than in more mature markets. In contrast, Elton et al. (2002) found that deviations for SPDRs in the U.S. market typically disappeared within the same trading day, reflecting highly efficient arbitrage and deeper liquidity. The lower speed of adjustment in India likely arises from higher transaction costs, limited arbitrage participation, and wider bid–ask spreads, which delay price convergence. Thus, while mean reversion occurs, it does so more gradually in the Indian context.

1.4 Comparison with Alternative Vehicles

1. Index Funds

An individual investor in India has a growing selection of Nifty 50 index funds to choose from. Among the most prominent are the UTI Nifty 50 Index Fund, Nippon India Index Fund – Nifty 50 Plan, and ICICI Prudential Nifty 50 Index Fund. When selecting among these, two key considerations emerge: tracking accuracy (how closely the fund replicates the Nifty 50 Total Return Index) and the magnitude of return shortfall due to expenses and operational drag. Tracking error is generally low across these funds, with R^2 values typically above 0.999, indicating near-identical movement with the benchmark. Differences in long-term performance are thus driven primarily by expense ratios and portfolio management efficiency.

Relative performance of a Nifty 50 index fund compared with the Nifty 50 Total Return Index (TRI) and Nippon BeES is influenced by several key factors. The first is costs. Index funds incur management fees and other operational expenses that reduce returns. Specifically:

- UTI Nifty 50 Index Fund: ∼17 basis points per annum
- Nippon India Index Fund – Nifty 50 Plan: ∼41 basis points
- ICICI Prudential Nifty 50 Index Fund: ∼36 basis points
- Nippon BeES: only 4 basis points

Thus, individual investors in Nippon BeES enjoy a significant cost advantage—paying 13 to 37 basis points less annually than in the comparable index funds.

Additionally, index funds face transaction costs whenever they buy or sell stocks due to inflows/outflows, index rebalancing, or dividend reinvestment. These cash-based trades incur brokerage, impact costs, and taxes. In contrast, Nippon BeES uses in-kind creation and redemption, where authorized participants deliver or receive actual stock baskets—virtually eliminating transaction drag for the ETF.

Another indirect cost for index funds is cash drag: they must hold small cash buffers to meet redemptions, slightly diluting returns. While some funds use Nifty futures to offset this, such flexibility is not applicable to ETFs like BeES, which maintain near-full investment at all times.

The second factor is index transition efficiency. Funds differ in how quickly and cost-effectively they adjust to Nifty 50 rebalances, inclusions/exclusions, or corporate actions (like bonus issues or mergers). Delays or poor execution can widen tracking error.

The third factor is securities lending. All three index funds actively lend portfolio stocks to short-sellers, generating extra income (typically 5–15 bps annually) that partially offsets expenses. Nippon BeES does not engage in securities lending, forgoing this revenue but maintaining structural purity and avoiding counterparty risk.

How do all these influences net out? Over the period 2013–2021, the UTI Nifty 50 Index Fund and ICICI Prudential Nifty 50 Index Fund outperformed Nippon BeES by 1.4 basis points per year (pre-tax), while the Nippon India Index Fund – Nifty 50 Plan delivered identical performance to BeES. Notably, Nippon BeES outperformed the Nifty 50 Total Return Index (TRI) by 5.7 basis points annually, despite its ultra-low 4 bps expense ratio—suggesting near-perfect tracking and minimal operational drag.

These differences are calculated pre-tax. When taxes are considered, another layer emerges. Capital gains taxes arise when funds realize gains—typically during index rebalancing (additions/deletions) or, for mutual funds, during net redemptions. Index funds may hold cash to manage outflows, often selling recently purchased (higher-cost) shares to minimize gains. However, Nippon BeES enjoys a structural tax advantage through in-kind creation and redemption: when authorized participants redeem units, they receive actual stocks—a non-taxable event. This allows the ETF to systematically deliver low-cost-basis shares to redeemers, keeping the portfolio's cost basis high and close to market value. As a result, capital gains triggered by index changes are minimized or eliminated in BeES, whereas index funds must sell shares in cash, potentially realizing taxable gains.

Thus, while the three index funds slightly outperformed BeES pre-tax due to securities lending income, Nippon BeES holds a clear edge post-tax—especially for taxable investors—thanks to lower fees, zero transaction drag, and superior tax efficiency via in-kind mechanics.

A major difference between index funds and Nippon BeES is that BeES units can be traded intraday on the NSE and BSE, while index funds are transacted only at end-of-day NAV. What does this tell us about the value of immediacy?

Over 2013–2021, the UTI Nifty 50 Index Fund and ICICI Prudential Nifty 50 Index Fund outperformed Nippon BeES by 1.4 basis points per year (pre-tax), while the Nippon India Index Fund – Nifty 50 Plan matched BeES exactly. Both BeES and the index funds outperformed the Nifty 50 TRI, but BeES delivered 5.7 bps above the benchmark despite its ultra-low cost.

Thus, investors choosing Nippon BeES over the top-performing index funds (UTI or ICICI) sacrifice 1.4 basis points annually in return to gain intraday liquidity. This implies that sophisticated investors value immediacy at at least 1.4 basis points per year—a modest premium reflecting the ETF's exchange-traded flexibility, real-time pricing, and ability to hedge or rebalance instantly.

We say "at least" because:

- BeES trades involve bid-ask spreads and brokerage commissions, whereas index fund transactions occur at NAV with zero load (for direct plans).
- The true cost of immediacy is therefore higher than 1.4 bps when execution costs are included.

In short, Nippon BeES offers intraday access at a very low implied cost—making the value of immediacy in the Indian market remarkably affordable, especially compared to active trading strategies.

2. Futures

The other alternative to Nippon BeES is holding short-term money market instruments plus Nifty 50 futures contracts. If futures traded at theoretical fair value—i.e., spot price plus cost of carry (financing) minus expected dividends—this strategy would replicate the Nifty 50 Total Return Index (TRI) minus transaction costs. In practice, however, Nifty futures frequently deviate from fair value due to market sentiment, liquidity, or dividend uncertainty.

We estimate the implied Nifty 50 spot price embedded in the futures price using:

- Monthly closing futures prices (2013–2021),
- Actual dividend yield derived from the gap between Nifty 50 TRI and Price Return Index,
- 3-month Government of India Treasury Bill yield as the proxy for short-term risk-free rate (financing cost).

Over the sample period:

- Nifty 50 TRI annualized return: 14.7150%
- Nippon BeES annualized return: 14.1762%
- Synthetic futures + T-bill strategy (implicit return): 12.6787%

Thus, the futures-based strategy underperformed both BeES and the benchmark by ∼150 bps and ∼204 bps, respectively. This suggests that, on average, Nifty futures traded at a premium to fair value—implying a negative basis for long positions, possibly due to persistent retail buying pressure, roll cost drag, or over-optimism in dividend forecasts.

Despite this, a regression of daily BeES price changes (net of accumulations) against changes in the futures-implied Nifty level shows:

Metric	Value
R^2	0.9813
Slope (β)	0.9919
Intercept	–0.00028 (not significant)
t-stat (slope)	74.65

This confirms near-perfect co-movement between BeES and futures-implied index levels, with BeES exhibiting slight dampening (0.8% less sensitivity), consistent with expense ratio drag and microstructure friction.

Key Insights:

- Futures offered no return advantage—in fact, they lagged due to mispricing.
- Nippon BeES wins on accessibility: no margin calls, no rollovers, no forecasting, and fractional unit trading.
- Many institutions are restricted from holding derivatives; retail investors lack expertise in managing futures positions.
- The 3-month T-bill yield provided a clean, market-based proxy for funding cost, enhancing the robustness of fair-value estimation.

Thus, despite theoretical appeal, Nifty futures underperformed in practice. Nippon BeES remains the superior passive vehicle—delivering benchmark-like returns with simplicity, liquidity, and structural efficiency.

2 Determinants of Volume

An important feature of ETFs is that their trading volume may be influenced by market conditions, investor arbitrage activity, and price misalignments between the ETF and its underlying index. Elton et al. (2002) found that SPDR (Spider) ETF volume was significantly associated with both price deviations from NAV and general market volatility. In this section, we examine whether similar dynamics exist for the Nippon India ETF Nifty 50 BeES (BeES) using data from 2013 to 2021.

We focus on the following potential determinants of trading volume in BeES:

- The deviation between the ETF's market price and NAV, interpreted as a potential arbitrage signal.
- The volatility of the Nifty 50 index, capturing market uncertainty or trading opportunity.
- The volume in the Nifty 50 futures market, as a proxy for overall index-related trading interest.
- The relative share of ETF volume to index futures volume, as a measure of investor preference for ETFs over futures.

2.1 Price-NAV Deviation and Future ETF Volume

We first examine whether the deviation between BeES's NAV and trading price on day t predicts its trading volume on day $t+1$. This regression captures whether pricing inefficiencies drive arbitrage or investor activity.

$$\text{Volume}_{t+1} = \alpha + \beta \cdot (\text{Price}_t - \text{NAV}_t) + \varepsilon$$

- $R^2 = 0.415$
- Intercept = 33.2 million (t = 42.87)
- Slope = –3.75 million (t = –41.87)

This result shows a strong and statistically significant negative relationship between ETF premium (Price ¿ NAV) and next-day volume. When the ETF trades at a premium to NAV, volume declines; conversely, when the ETF trades at a discount, volume rises. This behavior is consistent with arbitrage-driven trading — investors are more likely to buy when the ETF is undervalued (relative to NAV), driving volume up.

2.2 Nifty Volatility and ETF Volume

Next, we test whether volatility in the Nifty 50 index influences BeES trading volume. We regress the ETF's next-day volume on same-day Nifty volatility:

$$\text{Volume}_{t+1} = \alpha + \beta \cdot \text{Volatility}_t + \varepsilon$$

- $R^2 = 0.093$
- Intercept = 96,979 (t = 1.84)
- Slope = 58.89 million (t = 15.89)

The slope is large, positive, and highly significant. This suggests that BeES volume rises during periods of high Nifty 50 volatility — likely due to increased trading interest and portfolio rebalancing among both institutional and retail investors.

2.3 Relative ETF–Futures Usage and Market Volatility

To assess whether investors substitute between ETFs and futures during volatile periods, we regress the volume share (BeES volume / Nifty futures volume) on Nifty volatility:

$$\frac{\text{ETF Volume}_t}{\text{Futures Volume}_t} = \alpha + \beta \cdot \text{Volatility}_t + \varepsilon$$

- $R^2 \approx 0.00009$
- Intercept = 0.0118 (t = 70.74)
- Slope = –1.06e–10 (t = –0.48)

The extremely low R^2 and insignificant slope indicate that market volatility does not shift investor preference between ETFs and futures — at least not measurably. ETF volume as a proportion of futures volume remains stable, regardless of volatility spikes.

2.4 ETF Price Deviation and Nifty Futures Volume

To test whether trading interest in the index as a whole (proxied by Nifty futures volume) explains pricing inefficiencies, we regress the ETF's price-NAV deviation on same-day Nifty futures volume:

- $R^2 = 0.0013$
- Intercept = 413.23 (t = 4.96)
- Slope = 1.82 (t = 1.80)

Results are similar for lags of 1–3 days. While the coefficients are statistically marginal, the very low R^2 suggests that market-wide index volume has almost no explanatory power for ETF mispricing.

2.5 Multivariate Model: Price Deviation and Volatility as Joint Predictors

Finally, we run a multiple regression of next-day ETF volume on both price deviation and Nifty volatility:

$$\text{Volume}_{t+1} = \alpha + \beta_1 \cdot (\text{NAV} - \text{Price})_t + \beta_2 \cdot \text{Volatility}_t + \varepsilon$$

- $R^2 = 0.110$
- Intercept = 30,743 (t = 0.58)

- Slope (Deviation) = 8,195 (t = 0.05)
- Slope (Volatility) = 64.47 million (t = 17.47)

This confirms that volatility alone is the dominant driver of volume, while the price deviation's effect is not significant when both variables are included. The effect of arbitrage signals may already be captured in the volatility environment.

Summary of Findings

- ETF volume is strongly driven by volatility in the Nifty 50 index.
- Price-NAV deviation predicts volume, suggesting arbitrage-related trading behavior.
- However, when both variables are included, volatility dominates.
- Investor preference between ETF and futures does not shift significantly with volatility.
- Futures volume does not explain pricing inefficiencies in BeES.

These results mirror the conclusions of Elton et al. (2002), who also found that ETF volume increases with market activity and that price deviations have predictive value, but not in isolation from volatility. In the Indian context, BeES appears to respond similarly to broad market conditions, reinforcing its role as a responsive, exchange-traded index vehicle.

3 Conclusion

This study set out to replicate the seminal analysis of the SPDR S&P 500 ETF (SPY) conducted by Elton et al. (2002), but in the Indian context using Nippon India ETF Nifty 50 BeES (BeES) as the focal instrument. We evaluated BeES across multiple dimensions—return performance, tracking error, price-NAV deviations, comparative efficiency against index mutual funds and futures, and volume determinants—using data from 2013 to 2021. Our findings affirm that Nifty BeES serves as a highly effective and low-cost vehicle for passive index replication in India. Despite operating in a different regulatory and market microstructure environment than the U.S., BeES closely tracks the Nifty 50 Total Return Index (TRI) with minimal long-run deviation. The average annual underperformance relative to the TRI was a negligible 0.53 basis points, of which 1.20 basis points were attributable to price-based tracking error. Interestingly, the remaining component was positive, reflecting effective dividend reinvestment and low expense drag—features that contrast favorably with the dividend and cost frictions observed in the original SPDR study. When compared to major Nifty 50 index mutual funds such as UTI, ICICI Prudential, and Nippon India, BeES demonstrated similarly high tracking accuracy, with regression betas near unity and R^2 values exceeding 0.99. Although UTI and ICICI outperformed BeES slightly on a pre-tax basis due to securities lending income, the structural advantages of ETFs—including lower fees, in-kind creation/redemption, and superior tax efficiency—position BeES as a robust long-term indexing solution, particularly for taxable investors. In our investigation of market pricing efficiency, we found that BeES trades very close to its NAV, with daily deviations exhibiting significant mean reversion. About 63% of any price-NAV deviation persisted to the following day, and nearly 40% of its variation could be explained by previous-day deviations—suggesting a well-functioning arbitrage mechanism, albeit with slightly slower adjustment than in the more liquid U.S. markets. We also examined alternative indexing vehicles and found that a synthetic futures-based strategy underperformed both BeES and the TRI by over 150 basis points annually, largely due to persistent basis mispricing, cost of carry inefficiencies, and roll-over friction. Despite their theoretical elegance, futures failed to outperform BeES in practice, reaffirming the ETF's comparative advantage in delivering benchmark-consistent returns without derivative complexity or margin risk. Finally, we assessed the drivers of trading activity in BeES. Consistent with expectations, market volatility had a strong positive influence on ETF volume, suggesting active trading during turbulent periods. Price-NAV deviations also predicted volume, although their effect became insignificant when controlling for volatility. Interestingly, ETF-to-futures volume ratios remained stable across volatility regimes, indicating no significant substitution effect between the two instruments. In conclusion, Nippon BeES fulfills its mandate as a cost-efficient, transparent, and highly liquid indexing instrument. Its performance and structural efficiency closely resemble the idealized behavior envisioned in the design of SPDRs, but with important improvements:

lower expense ratios, real-time dividend reinvestment, and simplified tax handling. While BeES does not eliminate all sources of return slippage, it delivers an effective vehicle for passive investors in India—one whose deviations from ideal benchmark behavior are economically insignificant and operationally well-managed. These findings support the growing institutional and retail adoption of ETFs in India and highlight the continued relevance of the ETF structure in facilitating broad, efficient market access.

References

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